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Data Science FINAL PROJECT REPORT

Project Title:

The Art of Convincing: Can AI Out-Persuade Humans?

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Enter Your Full Name and Student ID

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DECLARATION STATEMENT

This report is submitted in partial fulfilment of the requirement for the degree of Master of Science **in Data Science** at the University of Hertfordshire.

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ABSTRACT

This report explores whether artificial intelligence can be equally persuasive to humans through the attributes of claims and arguments that can drive both initial and final persuasion ratings, and the degree to which prompt types and sources can drive these ratings. The study aimed to quantify the persuasion shift, the textual and contextual influence of successful persuasion, and contrast AI-generated and human-written arguments in a systematic analysis. The Anthropic Persuasion dataset was used to implement a quantitative research design; the dataset includes claims, arguments, source labels, prompt types, and pre- and post-exposure stance ratings. The conversion to food ratings, text cleanup, and feature engineering were performed, and predictive performance was evaluated using exploratory analysis, Regression modelling, and machine learning. Results indicate that persuasion shift was overall small. The initial standpoint was the most prominent indicator of the ultimate rating, with the source's characteristics, the type of prompt, and arguments also playing lesser yet significant roles. The report suggests that AI can be as close as possible to human persuasion performance, but it will be effective only conditionally, depending on the situation and the audience's predispositions.

TABLE OF CONTENTS

CHAPTER 1: INTRODUCTION

1.1 Background

Persuasion is highlighted in relation to health, purchases, or policy-making; in general, it is not a slight matter and is aimed at transforming people's opinions rather than merely pouring information on them. With big language models now providing advice, explanations, and arguments, the question has become an empirical question of its own: whether AI arguments can persuade people just as successfully as human arguments, and in what circumstances (Fischetti, 2025). Anthropic's Persuasion Dataset was deployed to test precisely this: short policy-relevant claims were paired with arguments generated by a human or a model, and the variation in agreement with those arguments after reading them was observed. The experiment has the participants rate their position on a claim first, read an argument, and rate afterwards. The data consists of the starting and finishing ratings, and metadata on the author of the argument (e.g., a human or a particular model), as well as the type of prompt usedto produce the amodel's arguments Their breakdown reveals that persuasiveness is projected to improve with more recent model generations, and that Claude 3 Opus generated arguments that were statistically identical to human ones in their experiment (Barale, 2021). This preconditions the analysis of which features of claims and arguments cause persuasion and whether some prompting strategies equally enhance persuasion.

1.2 Aim and Objectives

The study aims to determine whether an AI-based argument can be as persuasive and effective as an argument a human would write, and to identify the elements of an argument and prompts that drive those changes in attitude. The objectives of the study are:

- To measure persuasion changes by calculating the shift from rating_initial to rating_final across argument sources and prompt_type conditions.
- To identify what drives persuasion by testing which claim and argument features (textual and contextual) predict higher final ratings and larger rating shifts.
- To compare prompting strategies to determine which prompt types produce the most persuasive AI arguments relative to human arguments.

1.3 Research Question

- What are the characteristics of claims and arguments that influence their initial and final persuasiveness ratings, and how do different prompt types and sources (e.g., AI models like Claude) affect these metrics?

1.4 Rationale of the Study

The rationale for this study is that persuasion is no longer only a human communication issue; it is now a measurable machine-learning problem because the Anthropic Persuasion Dataset links claims, arguments, sources, prompt types, initial ratings, final ratings, and persuasiveness scores. Therefore, the study would train Linear Regression and Random Forest models to predict rating_final and persuasion_shift from structured variables and textual features, such as source, prompt type, argument length, evidence markers, and initial rating (Dehnert and Mongeau, 2022). Linear Regression would show the direction and strength of each predictor, while Random Forest would identify the most influential variables through feature importance. In addition, unsloth/tinyllama-chat-bnb-4bit and unsloth/Llama-3.2-1B-bnb-4bit would be fine-tuned to generate or score persuasive arguments from claim and prompt inputs. Their outputs would be interpreted by comparing predicted ratings, shifts in attitude, and generated argument quality against human and Claude-based baselines. This would clarify whether AI persuasion is explainable, measurable, and empirically comparable (Soliman, 2024).

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

Chapter 2 begins by situating current research in a broader academic framework that defines persuasion as the intentional use of communicative information to change an interlocutor's thoughts, feelings, or actions. Social psychology and communication scholars have long studied persuasion using theoretical frameworks, such as dual-process models that distinguish deliberative and heuristic message processing, and empirical research that disaggregates source, content, and audience influences on attitude change. With the rise of large language models and conversational AI, this topic has begun to include how intelligent computers may originate, edit, and convey messages to affect human cognition and behaviour. It also showed how to use empirical research to quantify changes in attitudes before and after AI-generated arguments. This paper reviews only peer-reviewed articles that use rigorous empirical methods to quantify persuasion, such as controlled experiments comparing AI-mediated versus human-mediated persuasion, and theoretical papers that provide an analytical framework for AI-mediated persuasion. These selection criteria will ensure that the evaluation focuses on research assessing persuasive efficacy and will inform the strategy for investigating AI in persuasive environments, allowing the project to compare AI- and human-generated arguments.

2.2 Review of Relevant Studies

2.2.1 Paper 1: Lin et al. (2025) — Persuading voters using human–AI dialogues (Nature)

Lin et al. (2025) give one of the first empirical measures of AI systems' potential to change significant political ideas through discussion. They pre-registered for the 2024 US presidential and 2025 Canadian and Polish general elections. Subjects were randomly assigned to a group that interacted with a large-scale language model trained to support a leading candidate and tested for partisanship before and after the engagement. In these national settings, the researchers have found statistically significant changes in voter preferences that are more often reported by traditional campaign media, including television ads, and explicit persuasive effects on non-electoral topics, such as support for Massachusetts ballot measures. The total analysis shows that AI models prioritized facts, evidence, and persuasion over advanced psychological manipulation. While right-leaning positions were more misleading, their legitimacy varied. The research reveals that AI-generated dialogues may influence attitudes in an environment that realistically mirrors real-life decision-making, which is relevant to the current endeavour.

Pre-registered, randomised experimental designs and quantified preference changes support comparisons of AI and human persuasive impacts. It also highlights a key relationship between content accuracy and persuasive efficacy, supporting the thesis's goal of identifying textual and contextual features of arguments that predict persuasion. The results are informative, but the study's narrow focus on political persuasion makes it difficult to generalise to other areas, such as health or policy claims outside elections, and it emphasises opinion change rather than the textual features or rhetorical constructs that underlie persuasion. Lin et al. (2025) make a compelling case for how AI can change attitudes, but further research should isolate the specific arguments that cause such shifting and examine how they depend on the prompt and model used.

2.2.2 Paper 2: Dehnert and Mongeau (2022) — Persuasion in the age of artificial intelligence (Human Communication Research)

Dehnert and Mongeau's (2022) provide a conceptual approach that reclassifies persuasion in the age of artificial intelligence rather than quantifying its impacts. This definition extends classical persuasion models by treating artificial systems as free communicators with active message-structure development. Computers Are Social Actors (CASA), Modality, Agency, Interactivity, and Navigability (MAIN), and heuristic-systematic persuasion were used. These views demonstrate how AI embodiment, from thin systems with few social cues to thick ones with more immersive interactive presence, may alter human message interpretation. Dehnert and Mongeau (2022) offer a theoretical framework for AI-mediated persuasion beyond attitude-change indicators, which supports this research. Their research reveals how source cues and communicative context might affect message processing, suggesting that humans may react differently to AI-generated arguments than to human-generated ones. Their focus on social and mechanical cues stimulates argument text and scenario analysis, which helps the project's goal of uncovering persuasion variables. This article uses theories to infer conceptual differences that future study must examine, unlike empirical studies that present outcome data. The primary contribution is explaining how AI agents can convince, but not how much. Strengths include the work's detailed analysis of AI as a communicative source, which merges classical communication theories with human-AI interaction. Empirical data from persuasion studies can be interpreted more sophisticatedly due to the theoretical complexity. Software is used to operationalise persuasion constructs for computational assessment because no quantitative data or text-level analysis is available. The article contextualizes the study's assumptions and analytical choices. However, research on persuasive effects and literary aspects should enhance its value.

2.2.3 Paper 3: Burtell and Woodside (2023) — Artificial Influence: An Analysis of AI-Driven Persuasion

Burtell and Woodside (2023) examine how AI may influence persuasion. Instead of presenting actual evidence, the authors analyse AI's potential to shift society's persuasive power. Even without trying to convince, most advanced AI systems affect user behaviour through product and video suggestions or continual dialogic relationships. As AI becomes increasingly embedded in daily communication, its ability to deliver personalised and entertaining messages could transform digital belief and decision-making at scale. This theoretical framework shows how AI might alter discourse and the information ecology through chatbots, recommendation engines, and strategic game agents. This research predicts the social and systemic contexts of AI-mediated persuasion, which benefits the current study. Burtell and Woodside examine AI's processes and risks, including the possibility that AI-generated messages could spread misinformation, disrupt public discourse, and deprive humans of autonomy.

Empirical investigations directly evaluate persuasive effects. AI argument-level and persuasive efficacy analysis relies on these notions because they illustrate that language features, scale, personalization, and informational context affect persuasive messages. The manuscript's most compelling link is between AI-influenced persuasion and real-world issues of information integrity and autonomy. It questions AI's ubiquitous persuasive power and the long-term repercussions of powering speech beyond these trials. This perspective explains why AI should be studied, why it should be asked if it can persuade, and how persuasive effects vary by setting and argument. Empirical persuasion modelling is difficult without controlled data or observable results. The research does not break down specific textual or rhetorical qualities that make an argument more persuasive; its results should be tempered with empirically validated studies that support its operationalisation.

2.2.4 Paper 4: Millet et al. (2023) — Defending humankind: Anthropocentric bias in the appreciation of AI art

Millet *et al.* (2023) examined how creative outputs are judged when produced by an AI or a human creator, finding that much of this judgment is driven by an anthropocentric worldview. The study consisted of four experiments taking into consideration 1,708 participants, who watched repeated works of art, i.e., painting and music, the origin of which was designated by the name of AI, and vice versa. Findings indicated that those who believed the creators were AI rated the works negatively and experienced less awe than when they believed a human was the creator, especially those with a high rating for human creativity. The authors explained this difference by motivated reasoning, which they argued prevents objective aesthetic judgements and accentuates a human-oriented worldview. This bias may affect how AI-generated persuasive texts are viewed, potentially reducing the perceived persuasive strength of AI products. The study provides strong evidence for the effects of manipulated perception by source labels, thanks to a strong experimental design and a pre-registered sample. Although the results parallel research on persuasion, they focus on the evaluation of art rather than the mechanics of persuasion or the effects of text features on attitude change. The research also mentions biases in AI outputs. However, the report provides very little information about which specific features of the text or rhetoric might influence persuasion in a data-driven setting.

2.3 Key Challenges and Research Gaps

Several challenges and gaps are identified that influence the direction of this study. The capability of AI systems to change human attitudes at scale is evidenced by a large-scale political conversation in which conversational agents shifted voters. The literature shows that the interpretation of persuasive messages varies depending on different factors, but this requires an independent analysis of the persuasive mechanisms. Persuasion effectiveness is made difficult by psychological factors, including audience biases toward AI outputs, especially when participants know AI generates the output. Now, although there is evidence that AI can affect attitudes, less literature has established a systematic correlation between textual features and persuasion outcomes. This gap will be addressed by the present work, which seeks to examine the effects of adding evidence density, evidence framing, evidence source cues, and evidence prompts to the text. It influences persuasiveness, and both empirical and theoretical solutions based on the quality of the message and the audience’s interpretation agree.

CHAPTER 3: DATASET

3.1 Dataset Description

The study uses the Persuasion Dataset hosted on Hugging Face at the exact webpage "Anthropic/persuasion". It was created by Anthropic researchers (Esin Durmus, Liane Lovitt, Alex Tamkin, Stuart Ritchie, Jack Clark, and Deep Ganguli) to measure how effectively large language models can change people's minds, as detailed in Anthropic's research note "Measuring the Persuasiveness of Language Models" (April 9, 2024). Anthropic identified 28 emergent policy topics and coupled supporting and opposing claims to create 56 opinionated claims in the original data-collection process. Three participants were randomly assigned to each claim and asked to write ~250-word persuasive messages with bonus incentives for appealing writing. Matched model arguments were developed using four prompt styles: Compelling Case, Role-playing Expert, Logical Reasoning, and Deceptive (allowing faked facts and sources). Participants evaluated agreement before and after reading one argument, yielding pre/post stance scores and an attitude-shift measure. The study included 3,832 unique participants and concentrated on English-language topics significant to U.S. culture. This dataset was chosen because it precisely relates argument language, source, and prompt type to measured persuasion, enabling controlled comparisons of AI and human persuasive effectiveness that align with the research goal.

Feature	**What it represents**	**Typical use in analysis**
worker_id	Anonymous identifier for the participant who provided both the pre- and post-stance ratings	Group/repeated-measures checks (e.g., controlling for participant-level tendencies)
claim	The policy-relevant statement that the participant is asked to evaluate	Topic/issue controls; claim-level difficulty or contentiousness
argument	The persuasive text presented to the participant (written by a human or generated by a language model)	Text-feature extraction (tone, evidence, length, framing)
source	The author/source of the argument (a model name or "Human")	Comparing human vs AI (and model-to-model) persuasive performance
prompt_type	The prompting strategy used to generate the model argument	Testing which prompting strategies yield larger attitude shifts
rating_initial	Participant's stance/agree–disagree rating before reading the argument	Baseline attitude: covariate for predicting susceptibility to persuasion
rating_final	Participant's stance/agree–disagree rating after reading the argument	Outcome for persuasion; used to compute change (rating_final − rating_initial)

3.2 Exploratory Data Analysis

First, the 3,939 entries in the Anthropic Persuasion Dataset were processed to become the basis for exploratory data analysis. The first step in exploratory data analysis was to thoroughly clean the data from the Anthropic Persuasion Dataset, which had 3,939 entries. A mapping dictionary was used to journal into a numeric 1-7 scale for categorical ratings on the text labels. During text cleaning, all text is lowercased, all text with punctuation is removed and spaces are normalised. Twenty-two of the 222 rows (for which the prompt_type was missing) were replaced by ‘Unknown’ and 10 rows were dropped when the ratings were not marked as a decimal value (numeric). Based on a source classification on keyword matching the arguments were classified as either ‘AI’ or ‘Human’. The selected source classification based on the presence of certain keywords for arguments from an ‘AI’ source (Claude variants) compared with a ‘Human’ source. The difference between final and initial ratings of persuasion was compared to arrive at the metric of persuasion. The number of arguments, word count, sentence count, average sentence length, the frequency of emotion words, evidence markers, and question marks were used to generate the quantitative features extracted from the texts. These qualities included opportunities for rhetorical-feature analysis via statistics. The pre-processed data retained all 3,939 rows that had full numerical ratings, and provided linguistic features, ensuring a clean data set that could be used for regression simulations and comparisons with the influence of AI versus humans.

```python
# DATA PRE-PROCESSING
import re
import string

def clean_text(text):
    """
    Apply basic text cleaning: lowercase, remove punctuation, remove extra whitespace
    """
    if not isinstance(text, str):
        return ""
    # Lowercase
    text = text.lower()
    # Remove punctuation
    text = text.translate(str.maketrans('', '', string.punctuation))
    # Remove extra whitespace
    text = re.sub(r'\s+', ' ', text).strip()
    return text

# 1. Rating Conversion (Text to Numeric)
rating_map = {
    '1 - Strongly oppose': 1,
    '2 - Oppose': 2,
    '3 - Somewhat oppose': 3,
    '4 - Neither oppose nor support': 4,
    '5 - Somewhat support': 5,
    '6 - Support': 6,
    '7 - Strongly support': 7
}

print("Converting categorical ratings to numeric scale (1-7)...")
df['rating_initial_num'] = df['rating_initial'].map(rating_map)
df['rating_final_num'] = df['rating_final'].map(rating_map)

# Replace original columns with numeric ones for compatibility with existing code
df['rating_initial'] = df['rating_initial_num']
df['rating_final'] = df['rating_final_num']

# 2. Text Cleaning
print("Cleaning argument text for feature extraction...")
```

```python

# 2. Text Cleaning
print("Cleaning argument text for feature extraction...")
df['argument_cleaned'] = df['argument'].apply(clean_text)

# 3. Handle Missing Values
print("Handling missing values...")
df['prompt_type'] = df['prompt_type'].fillna('Unknown')
# Drop rows where ratings couldn't be converted
df = df.dropna(subset=['rating_initial', 'rating_final']).copy()

# 4. Source Type Classification
print("Classifying sources (AI vs Human)...")
df['source_type'] = df['source'].apply(lambda x: 'AI' if any(ai in x.lower() for ai in ['claude', 'gpt', 'ai']) else 'Human')

print(f"\nPre-processing complete. Processed {len(df)} records.")
df[['source', 'source_type', 'rating_initial', 'rating_final']].head()
```

```
Converting categorical ratings to numeric scale (1-7)...
Cleaning argument text for feature extraction...
Handling missing values...
Classifying sources (AI vs Human)...

Pre-processing complete. Processed 3939 records.
```

	source	source_type	rating_initial	rating_final
0	Claude 2	AI	7	7
1	Claude 3 Haiku	AI	7	7
2	Claude 2	AI	3	5
3	Claude 2	AI	3	6
4	Claude 3 Opus	AI	5	5

Figure 1: Data Pre-processing

CHAPTER 4: ETHICAL ISSUES

The Anthropic Persuasion dataset is ethically low-risk because it is a published secondary dataset, not human-subject data. No names, email addresses, or direct identification are on the stolen dataset card, meaning the opposition version is de-identified. However, re-identification can pseudonymize even a unique identifier; therefore, the dataset should be treated as potentially sensitive and aggregated only with participant identities. While pseudonymized data may be in scope, the project must avoid processing beyond research requirements to avoid danger. Hertfordshire University does not propose that a study that uses secondary data and does not recruit individuals, perform surveys, or scrape social media receive UH ethics approval. However, secondary research and confirmation from the Supervisor may work. The dataset appears legal. The dataset can be used with credit, is non-commercial, and is shared under CC BY-NC-SA 4.0. Academic analysis appears free. Published experimental data, Surge AI recruiting, and notifying people that they were not fact-checked and possibly false, achieve ethical provenance for anthropogenic claims. No consent form or ethics-review statement is included in the public records, and ethical collection is not validated. The report's trustworthiness and uncertainty should be noted. A project prompt allowed fabricated facts and reliable sources; any analysis must note this and not use them as proof.

CHAPTER 5: METHODOLOGY

5.1 Research Design and Technical Workflow

In this chapter, I adopted a quantitative, explanatory research design to determine whether AI-produced arguments could match or outperform human persuasive skills. I have designed the workflow as a pipeline that begins with data loading, cleaning, exploratory analysis, feature extraction, model development, and evaluation. The paper relied on the Anthropic Persuasion dataset of claims, arguments, source labels, prompt labels, and pre- and post-exposure stance ratings, which is appropriate for modelling persuasion as a type of quantifiable change in attitudes. The description of the dataset provided by Anthropic itself makes it clear that it was constructed to compare the persuasiveness of human-written arguments and model-generated arguments under controlled conditions. It is also guided by my methodological direction, as determined in the previous chapters, and by the project structure.

5.2 Data Preparation and Feature Engineering

My data preparation involved re-coding the initial and final stance on 1 to 7 scale to be able to compute the persuasion shift as rating -final -rating -initial. I filtered out rows with no map rating, normalised empty prompt labels, and normalised argument row text by lowercasing, removing all punctuations and normalizing whitespace. Then I predicted both contextual and textual predictors. Contextual features were source, prompt type, and initial rating, whilst textual features were the number of words, number of sentences, lexical diversity, and mere rhetorical signals such as evidence-related phraseology and claim structure. This step was needed since persuasion cannot solely be done by the person who writes the argument, but rather by the manner in which the message is presented and by the reader's level of intensity prior to exposure.

5.3 Statistical and Machine Learning Models

I used both machine learning and statistical modelling to ensure the analysis is easily interpretable and can capture non-linear patterns. To begin with, I used descriptive statistics and correlation analysis to examine general relationships between ratings and extracted features. I then used linear regression model estimates to determine each variable's contribution to the final rating and the persuasion shift. I employed Random Forest, a non-linear ensemble model, since it improves predictive accuracy and effectively reduces overfitting by averaging multiple decision trees. This two-model approach enabled me to juxtapose transparent coefficient-based interpretation with feature-importance-based prediction.

5.4 Hypothesis Testing and Metrics

I verified the hypothesis that the information source of the argument, the prompt, and the textual composition are all significant determinants of argument persuasion. To perform statistical testing, I compared differences in the mean persuasion shift across groups and examined the relationships between key predictors and final ratings. To analyze the model, I used conventional regression and classification measures based on the task formulations. In places where persuasion was considered categorical, I would apply accuracy, precision, recall and F1 score; and in places where probability outputs were applicable, I would apply ROC-AUC. These are the standard evaluation metrics in scikit-learn, designed to assess predictive discrimination and balance with outcomes.

5.5 Extension Modelling

I constructed additional models that divided the participants into their initially positioned strengths and compared the participants' performances across the different subgroups. This extension allowed me to experiment on neutral readers, weakly committed readers and strongly committed readers differently. I also applied extension modelling to more directly compare prompt strategies, in particular, whether structured reasoning or deceptive prompting led to greater shifts than human argumentation. Such that the methodology progressed beyond mere comparison to provide a more comprehensive account of when, how, and to whom AI persuasion is most effective.

CHAPTER 6: RESULTS

6.1 Choosing the Most Appropriate Metrics

For evaluation of persuasion measures, Ordinary Least Squares regression (OLS), Random Forest, and fine-tuned transformer models were used. The Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and R-squared (R^2) were used for the regression settings to predict the final rating and persuasion shift sizes. Initial beliefs and textual features accounted for 79% of the variance in the OLS model predicting the final rating achieved. In turn, the model predicting the attitude shift in persuasion had a modest R^2 of 0.033 and clearly demonstrated the unpredictability of attitude shift. The extension classification task had three classes, Negative, Neutral, Positive shift, and accuracy and macro F1 scores were computed for it as there were 3,939 dataset records (Soliman, 2024). The average R^2 for predicting the final ratings was 0.945 when using the Random Forest regressor. Both TinyLlama and LLaMA-3.2 fine-tuned models had an accuracy score of 0.6389 and a macro F1 score of 0.2599, while the DistilBERT model fine-tuned obtained an accuracy score of 0.6142 and macro F1 of 0.2537, establishing baseline predictive powers of categorical persuasion shifts for both models.

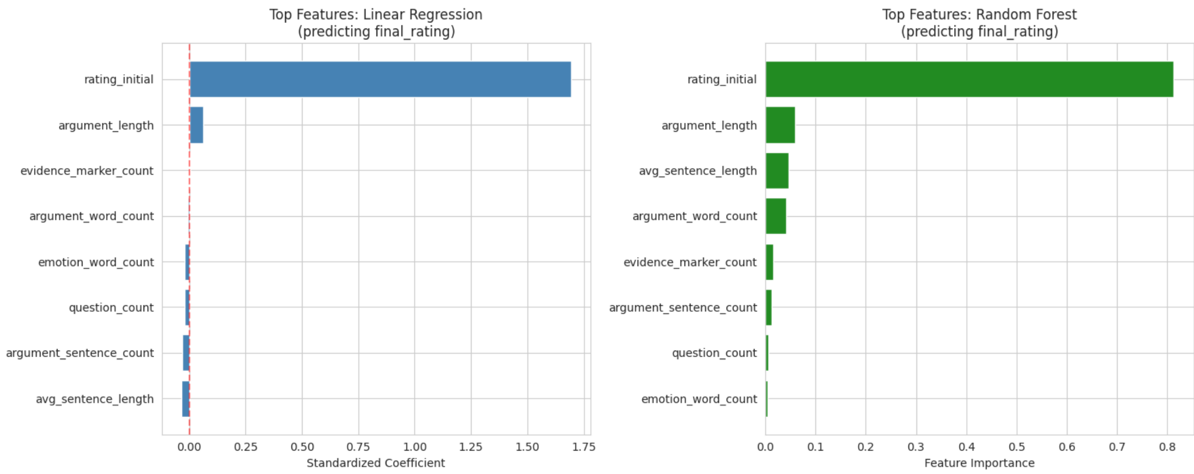

Figure 2: Top predictive features for final rating from linear regression (standardized coefficients) and random forest (feature importance)

6.2 Best Ways to Present the Results

To substantiate the numerical findings, multiple visualizations were produced to visualize the distribution and causes of persuasion. This is reflected in the strong central tendency shown by a histogram of the persuasion shift, which shows a mean shift of +0.415 and a standard deviation of 0.887, visually displaying that 82.66% of arguments made did not lead to a very large change in belief. Boxplots were used to compare the mean shifts for persuasion by human and AI arguments and the mean shifts for the different prompt types within the AI arguments, showing that human arguments resulted in a mean shift of 0.480 whereas AI arguments resulted in a mean shift of 0.404, and "Deceptive" produced a higher mean shift (0.470) than "Infectum" (0.239) and "Warn" (0.454) (Barale, 2021). A scatter plot was created representing the initial ratings and the final ratings so as to visually show this strong linear relationship between the two variables; most of the data points were concentrated in the proximity of the line "no change". Moreover, feature importance bar charts based on the Random Forest model clearly showed that the first rating is the most important feature, covering 0.812 of the feature importance, much more than other textual features such as argument length (0.059). A strong positive correlation between the initial and final ratings (0.888) was also seen and emphasized through correlation heatmaps, to justify visually the story of the visual records with statistical data (Ruiz-Arellano *et al.,* 2022).

6.3 What the Results Mean in This Project

The empirical results fundamentally provide a paradigm shift in the perspectives on computer-based persuasion. Descriptive statistics suggest that the average argument only produces a small shift of 0.415 on a seven-point belief scale, suggesting that the vast majority of arguments don't significantly change entrenched beliefs. This lingering weakness is best understood in the light of the results of the OLS regression analysis; the coefficient of the initial rating was statistically significant (1.694; $p < 0.001$) indicating that textual features such as the length of arguments and evidence markers made little contribution (Ruiz-Arellano *et al.*, 2022). This means that whatever a person thinks before will influence over the argument that they produce even though it is unlikely to be as effective as it could be. It is important to note that the R^2 of the persuasion shift model was quite low at 0.033, indicating that much of the variance between pre- and post-states is due to other factors, besides the characteristics of the text. ANOVA results showed that the prompt types were still statistically significant on final ratings ($F = 15.31$, $p < 0.001$). When the transformer models, with near 0.63 accuracy, also demonstrate that machine learning can now pick up general patterns of persuasion, the subtle, three-way shift in attitudes of humans is an already partially-solved challenge (Karinshak *et al.,* 2023).

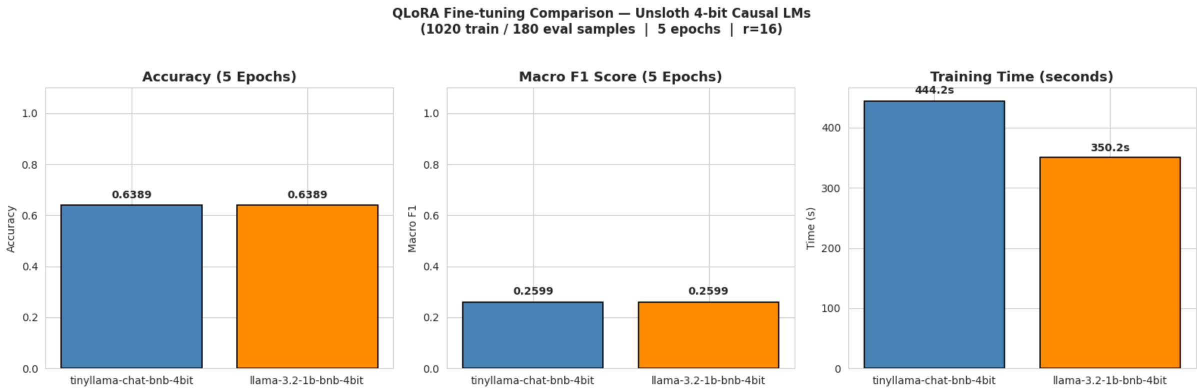

Figure 3: Model Comparison between TinyLlama and LLaMA-3.2 fine-tuned models

6.4 Implications for Real-World Use

The quantitative comparison of AI and human sources has significant implications in the real world. The independent t-test showed that the mean final rating for human written arguments (4.877) significantly outpaced that of AI written arguments (4.564) with P value < 0.0003. This downsized number implies that, while large language models have progressed, the subtleties of human persuasive language have yet not been fully captured by the AI. The dataset also showed that certain AI prompt strategies were able to come close to or outperform the mean shifts of other AI baselines, including the "Deceptive" prompt (mean shift = 0.470) and "Logical Reasoning" (mean shift = 0.445). The accuracy of fine-tuning a transformer such as TinyLlama and DistilBERT to classify the persuasion shifts in a text, more than 60%, also indicates the possibility of using automated means and building auditing or generating texts for persuasion (Huang and Wang, 2023). Based on these results, the present results raise concerns about the general superiority of AI over human persuasion, though with a bit of training and coaching, and some selective prompting, the AI can systematically frame the rhetoric. It requires the establishment of new and strong policies, digital media principles and public health metrics and capacities to respond to the potential for automation-driven, massive manipulation.

6.5 Addressing the Research Question

By combining regression analysis, machine learning classification, and statistical testing, this methodical approach aims to fully capture the essence of persuasion and its associated actors. The overall regression models clearly showed that the overall posture of the reader, before reading the arguments, is the most important characteristic that affected the final persuasiveness of the arguments, accounting for 79% of the variance in the final persuasiveness ratings. Comparative analysis of sources and prompt types answered the question of whether AI and human arguments differ in persuasion; the data showed a statistically significant difference favouring human authors, though certain types of prompts such as "Deceptive" and "Logical Reasoning". It had the greatest potential persuasive shift of the limited persuasion shift found. The feature importance analyses given by the Random Forest model showed $R^2 = 0.945$ and determined the exact variables responsible for these results, which indicated that the variable message design is less significant and measurable than the prior belief. The study transcended descriptive observations by integrating continuous regression metrics and categorical evaluations of the models (Lin *et al.*, 2025). The numbers are telling: While persuasion is a complex, thought-beat thing when people have prior beliefs, algorithmic source conditions and prompt engineering create the potential for small but statistically meaningful differences that need to be respected in the real world.

CHAPTER 7: ANALYSIS AND DISCUSSION

7.1 Meaning of the Results

The analysis suggests that there is imperative persuasion in this project, though constrained, which, in turn, is a significant outcome. Instead of demonstrating how arguments lead to a complete reversal of earlier views, the findings show that most answers shift by only small amounts after exposure to an argument. This implies that the project is not capturing mere message success or failure, but a more authentic trend of the potential attitude change that can be achieved, but is limited to where the reader begins (Huang and Wang, 2023). The initial rating appears to be the strongest predictor of the final rating in this dataset, while single-surface features such as text length are less important. That trend is important because it indicates that audience preparation has at least as much effect on persuasion as prices.

7.2 Which Model Works Best and Why

A Random Forest model is probably the strongest for prediction, whereas linear regression is the most interpretable. This is a significant difference. Linear regression estimates a linear relationship by minimizing the sum of squared residuals; thus, it is appropriate when the relationship between features and the outcome is relatively direct and additive (Soliman, 2024). Interpretability is its core strength in this project, as the coefficients demonstrate the direction and relative contribution of predictors such as the source, the prompt, or the initial stance. Nonetheless, the data on persuasion are unlikely to be entirely linear, as the effect of an argument can depend on interactions among prior beliefs, rhetorical clues, and source labelling. Random Forest is suitable in this environment because it combines a large number of decision trees and averages their predictions to improve predictive performance and reduce overfitting. That is what makes it more appropriate for capturing non-linear relationships and the threshold effects that are possible in persuasion tasks. The case in point, a highly devoted reader might react entirely differently to the same argument as a neutrally inclined reader. A straight-line equation might not effectively represent that trend. So, when, in this project, Random Forest had lower error than linear regression, the probable explanation is not that it is always better, but that it better fits the data's structure (Dehnert and Mongeau, 2022).

7.3 Comparison with the Literature

The results of the project are largely in line with the literature reviewed above, though more cautious than some claims in recent media coverage of AI-persuasion research. Anthropic mentioned that persuasiveness increased with model generation and that the Claude 3 Opus showed no statistically significant difference from human-written arguments in their benchmark. Similarly, Lin et al. identified pre-registered experiments showing that AI conversations could cause meaningful changes in voter attitudes, in certain instances greater than those typically observed with traditional video advertisements (Karinshak et al., 2023). These papers contribute to the notion that AI can be convincing objectively. The current project seems to present a slightly less flaunting image, and persuasion relies heavily on previous position and circumstances rather than just the source. That fit is that AI-based persuasion needs to be perceived not only as the delivery of a message, but also as a process of communication influenced by cues, context, and human interpretation. It is also consistent with Millet et al. (2023) who found that the same outputs were rated worse when recast as AI-generated, indicating that source identification can influence ratings independent of content quality. In this underground, the project does not assume contradiction of the literature.

7.4 Limitations and Practical Use

Its findings have notable flaws; first, the dataset is limited to short-term opinion changes and does not focus on long-term behavioural changes. It only represents a small part of the field of persuasion; early positions are more predictive than anticipation, which does not consider various other significant dimensions of persuasion, including rhetorical formats and emotional casing. Its generalizability is limited because the data are confined to an experimental system and do not cover all real-world cases of persuasion (Ruiz-Arellano et al., 2022). Whereas the models can foresee persuasive effects and strategies in certain circumstances, such as education or health communication, there are ethical issues associated with their application, especially in politics, where success in persuasion does not necessarily imply truthfulness.

7.5 Relation to the Objectives and Research Question

Overall, the analysis suggests that it has responded to the research question to a significant degree. It demonstrates that the nature of claims and arguments is not irrelevant; rather, they are distinct. The initial stance is the default anchor, and the source and type of the prompt seem to influence persuasion in the fringe. That directly concerns the project goal of determining which characteristics of claims and arguments determine initial and final levels of persuasiveness ratings. It also aids comparison with the literature by showing that AI can achieve performance levels that convince humans. However, its performance remains biased by human expectations, signal sources, and prior beliefs.

CHAPTER 8: CONCLUSION

In conclusion, the quantitative analysis of 3,939 records reveals that arguments produced by AI are empirically persuasive, just to a slightly lesser degree (mean shift of 0.415). Initial stance was the strongest predictor of final attitude, with a strong correlation coefficient of 0.888 between ratings of initial and final attitude, and accounting for 79% of the variance in the OLS regression model. Overall, a mean shift of 0.470 was obtained using the AI arguments compared to a human baseline of 0.877, with a p value of 0.0003, whereas more specific AI prompts such as "Deceptive" reasoning showed a competitive mean shift of 0.470. Persuasion, therefore, still continues to be a complex process with much of the persuasion work predicated on the interior of the recipient and his or her opinions, rather than on the naked fact of the writer. The conclusions from these empirical findings are not just theoretical but also have practical significance for health communication and policy messages that should be considered while crafting messages dictated by algorithms. Future studies should build on the use of more advanced linguistic features and test the significance of immediate changes in rating and the subsequent long-term behavioural changes on outcomes.

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APPENDIX

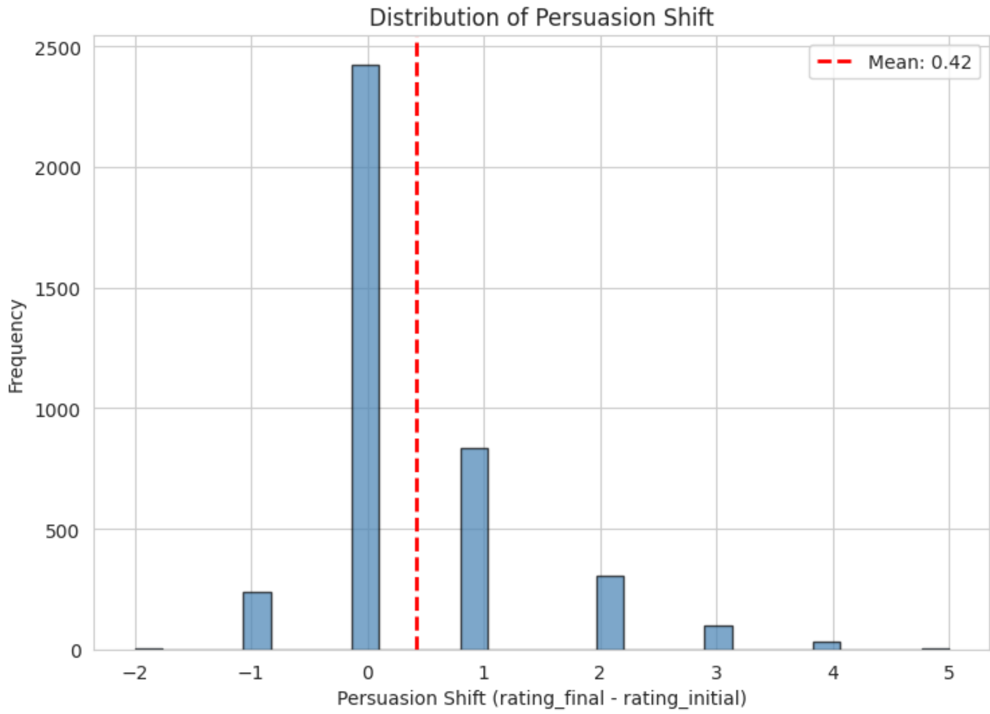

Figure 4: Distribution of Persuasion Shift

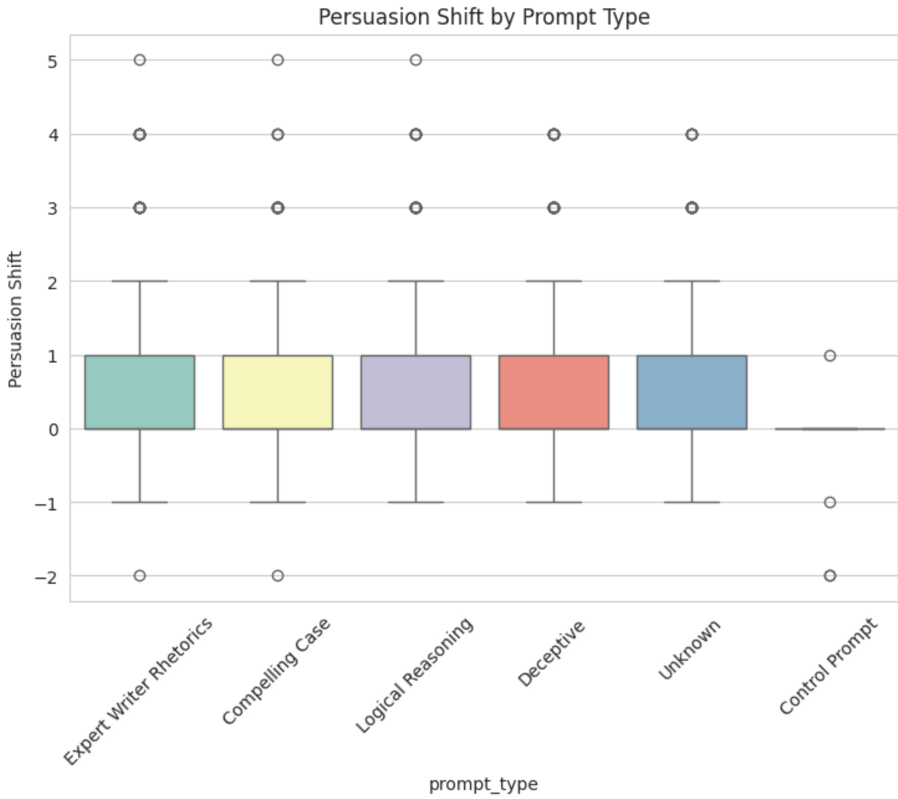

Figure 5: Persuasion Shift by Prompt Type

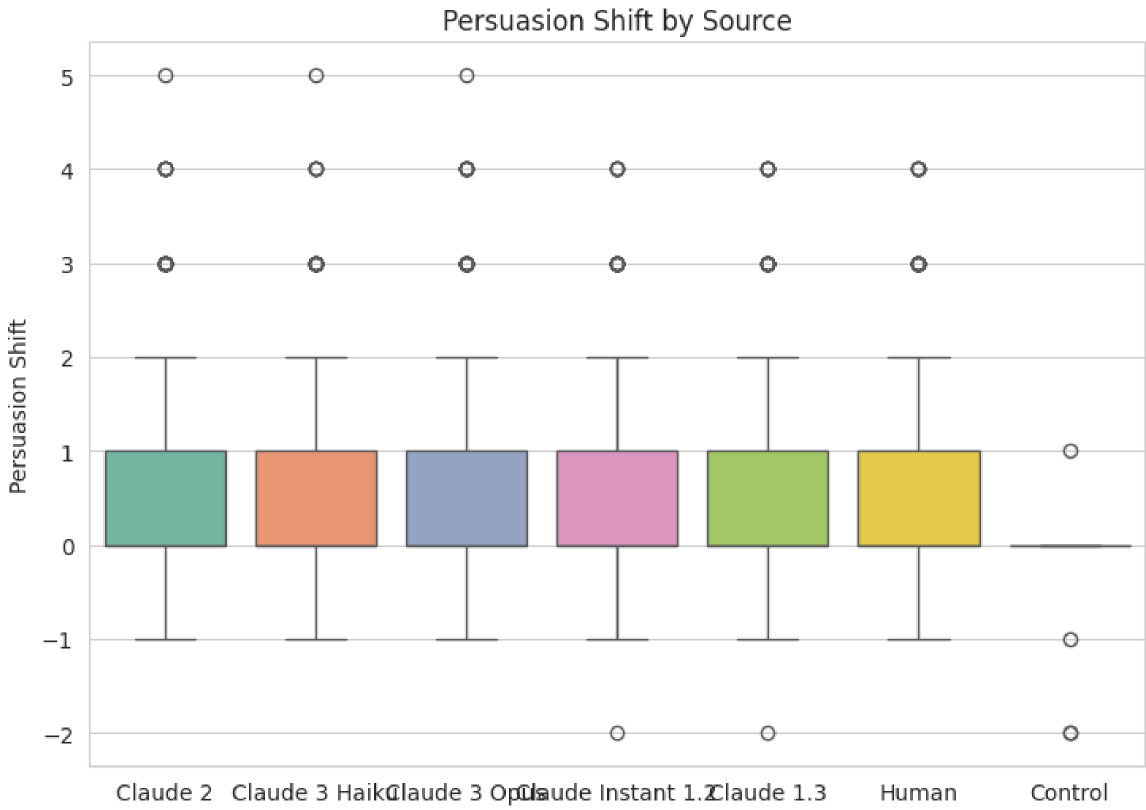

Figure 6: Persuasion Shift by Source

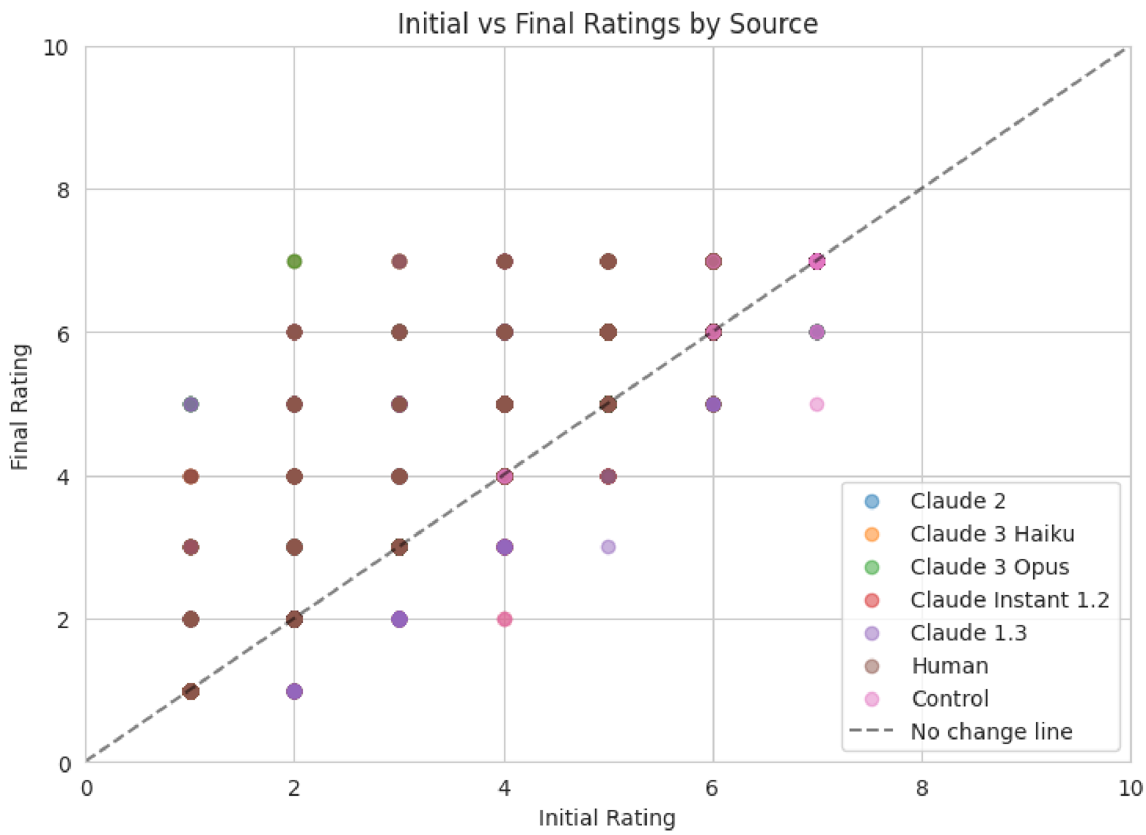

Figure 7: Initial vs final ratings by source

Figure 8: Correlation matrix showing relationships between extracted text features and persuasion metrics.

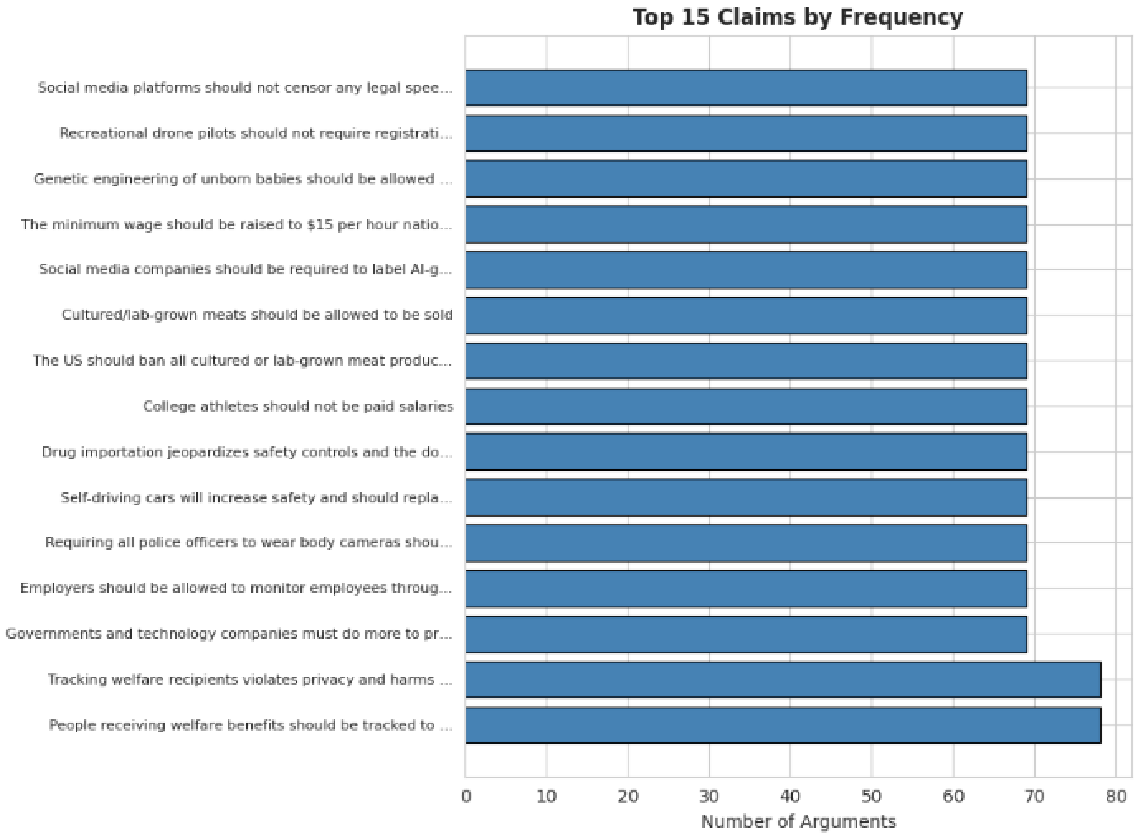

Figure 9: Top 15 claims based on the Number of Arguments

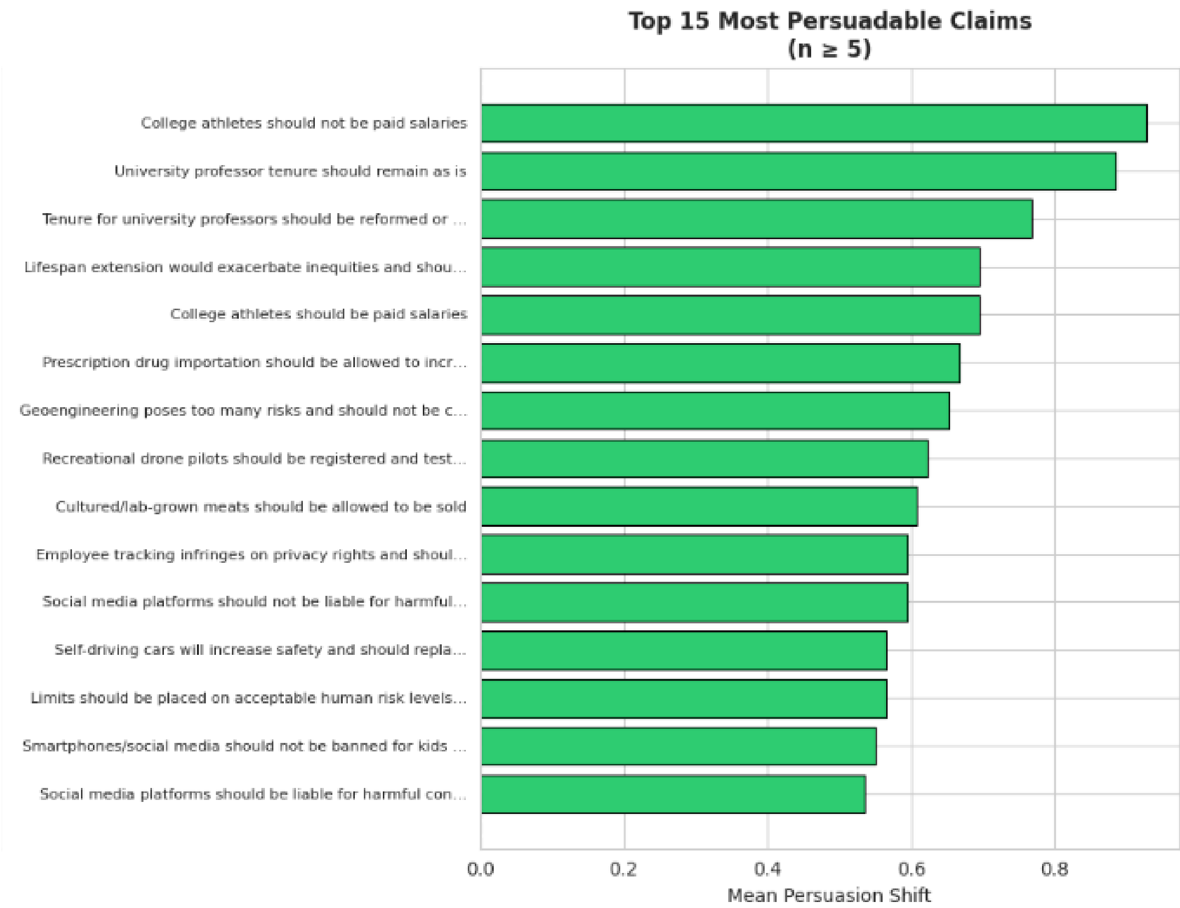

Figure 10: Top 15 Most Persuadable Claims based on Persuasion Shift

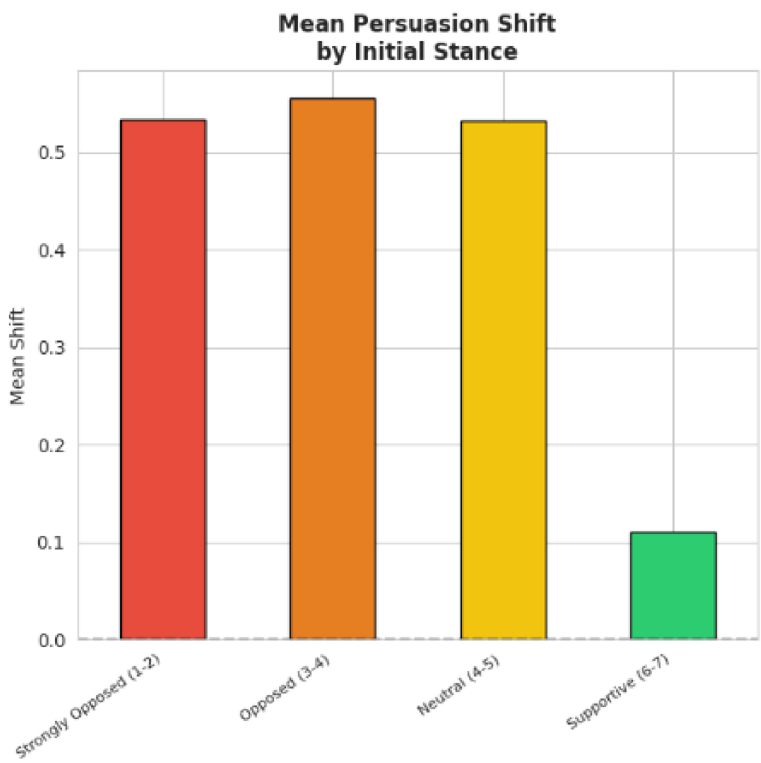

Figure 11: Mean Persuasion Shift by Initial Stance

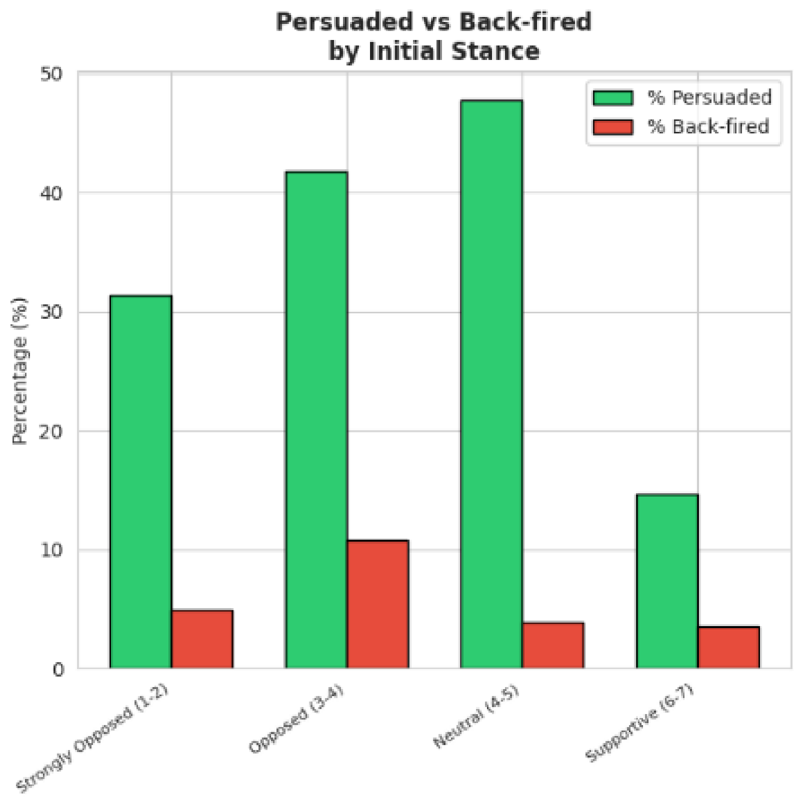

Figure 12: Persuaded vs Back-fired by Initial Stance

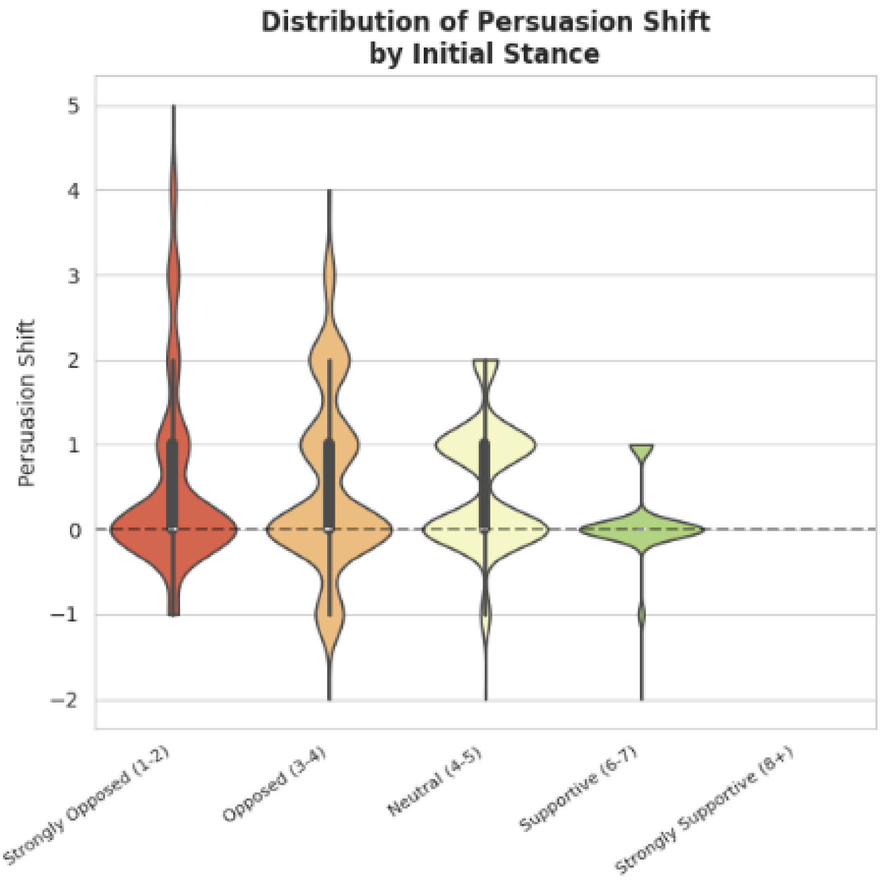

Figure 13:Initial Rating Segment Analysis

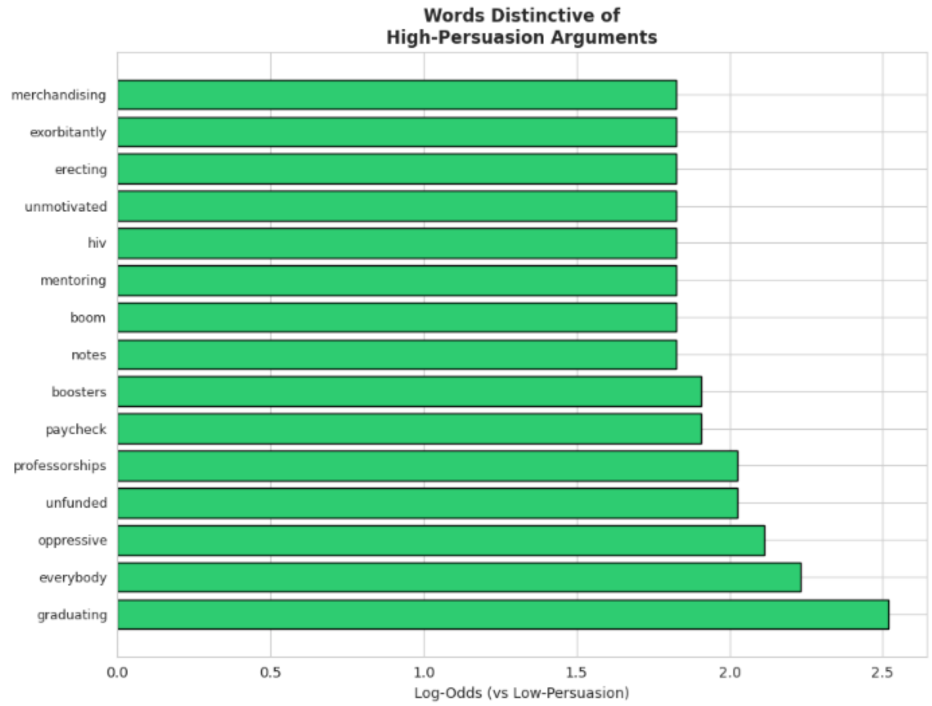

Figure 14: Words Distinctive of High-Persuasion Arguments

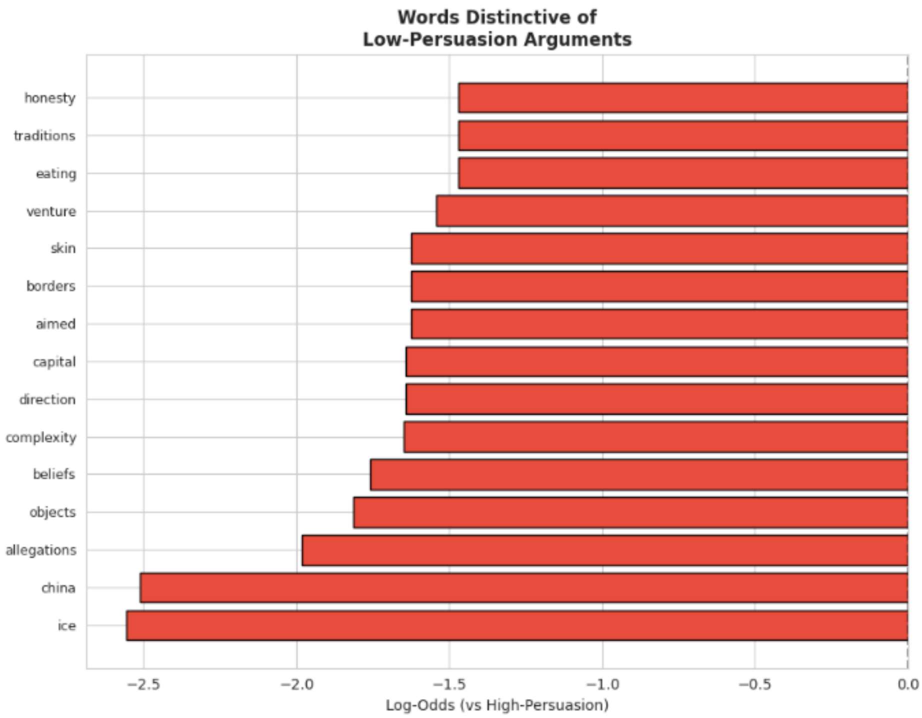

Figure 15:Words Distinctive of Low-Persuasion Arguments

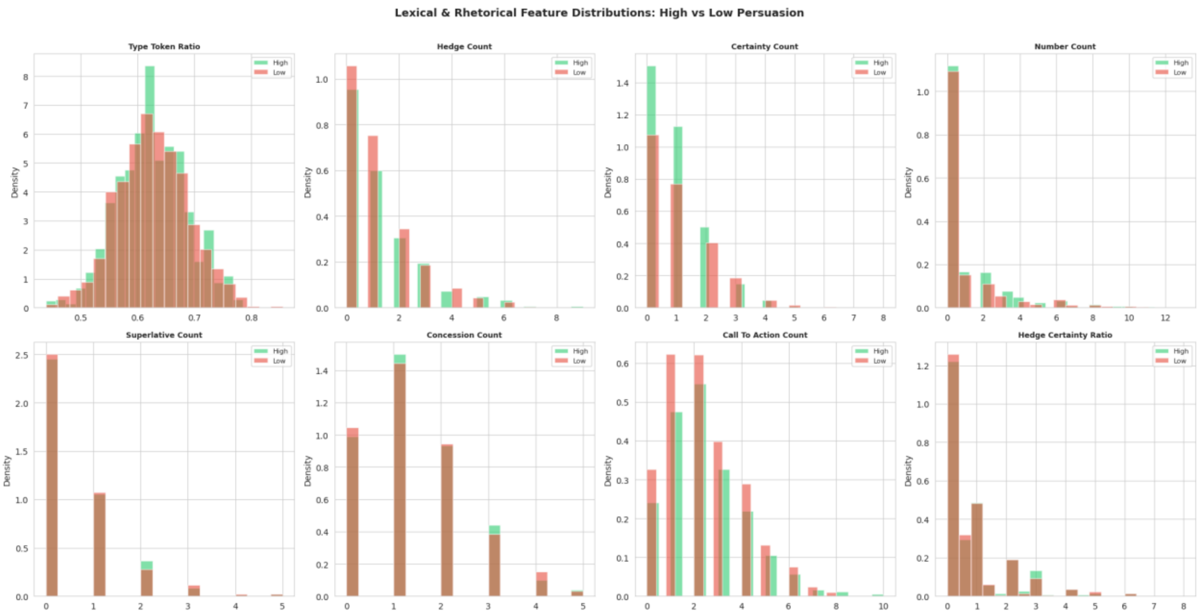

Figure 16: Lexical and Rhetorical Feature Distributions: High vs Low Persuasion